Supporting Information

Sodium Borohydride Reduction of Aryl Methyl Ketones: an Example Project for Chem SI Assistant

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Example project shipped with Chem SI Assistant — not a real Supporting Information. This example was made for Chem SI Assistant. The compounds and their NMR data, R_f values and physical states are from the cited literature; the spectra are simulated from those data. The procedures, the optimization study (Table S1), the yields, masses and HRMS found values are invented for illustration. Nothing here is a measurement.

Table of Contents

General Information **S3**
Reaction Optimization **S3**
Synthesis and Characterization **S4**
General Procedure A: Reduction of aryl methyl ketones with $NaBH_4$ S4
1-Phenylethan-1-ol (2a). S4
1-(4-Methoxyphenyl)ethan-1-ol (2b). S4
1-(4-Fluorophenyl)ethan-1-ol (2c). S5
1-(4-Chlorophenyl)ethan-1-ol (2d). S5
1-(4-Bromophenyl)ethan-1-ol (2e). S5
1-(Naphthalen-2-yl)ethan-1-ol (2f). S5
1-(4-Bromophenyl)ethyl acetate (3). S6
Spectra **S7**
References **S22**

General Information

About this example. This example was made for Chem SI Assistant. The compounds and their NMR data, R_f values and physical states are from the cited literature; the spectra are simulated from those data. The procedures, the optimization study (Table S1), the yields, masses and HRMS found values are invented for illustration. Nothing here is a measurement.

General methods

Unless otherwise noted, reactions were carried out under air in oven-dried glassware with magnetic stirring. Reagents and solvents were purchased from commercial suppliers and used as received. Reactions were monitored by thin-layer chromatography (TLC) on silica gel 60 F_{254} plates (visualized with UV light, 254 nm, and $KMnO_4$ stain). Flash column chromatography was performed on silica gel (40–63 μm).[1] The reduction conditions were optimized as shown in Table S1.

NMR spectroscopy

^1H, $^{13}C\{^1H\}$ and $^{19}F\{^1H\}$ NMR spectra were recorded at 298 K in $CDCl_3$ on 400, 500 and 600 MHz spectrometers (^1H frequencies; ^{13}C at 126 and 150 MHz, ^{19}F at 376 MHz). Chemical shifts (δ) are given in ppm, referenced to the residual solvent signals ($CHCl_3$: δ_H 7.26; $CDCl_3$: δ_C 77.16)[2] and, for ^{19}F, to external $CFCl_3$ (δ_F 0.00). Multiplicities are given as s (singlet), d (doublet), q (quartet), qd (quartet of doublets), m (multiplet) and br (broad); coupling constants (*J*) are given in Hz.

Mass spectrometry

High-resolution mass spectra (HRMS) were recorded with electrospray ionization (ESI) in positive-ion mode; the $[M+Na]^+$ ions are reported.

Abbreviations

Ac_2O, acetic anhydride; DMAP, 4-(dimethylamino)pyridine; rt, room temperature.

Reaction Optimization

The conditions of the reduction were optimized with 4′-bromoacetophenone, which gives alcohol **2e** (Table S1). Sodium borohydride (1.5 equiv) in methanol at 0 °C gave the highest yield (entry 1) and became General Procedure A. In THF the reduction was slow (entry 4), and sodium triacetoxyborohydride did not reduce the ketone (entry 7).

Table S1. Optimization of the reduction of 4′-bromoacetophenone

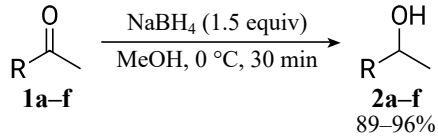
Entry[a]	Reductant (equiv)	Solvent	Temperature	Time	**2e** (%)[b]
1	$NaBH_4$ (1.5)	MeOH	0 °C	30 min	**95**
2	$NaBH_4$ (1.0)	MeOH	0 °C	30 min	81
3	$NaBH_4$ (1.5)	EtOH	0 °C	30 min	90
4	$NaBH_4$ (1.5)	THF	0 °C	30 min	23
5	$NaBH_4$ (1.5)	MeOH	rt	30 min	91
6	$LiBH_4$ (1.5)	THF	0 °C	1 h	88
7	$NaBH(OAc)_3$ (1.5)	CH_2Cl_2	rt	24 h	<5

[a] Reaction conditions: 4′-bromoacetophenone (0.50 mmol), reductant, solvent (2.5 mL).
[b] Isolated yield.

Synthesis and Characterization

General Procedure A: Reduction of aryl methyl ketones with $NaBH_4$

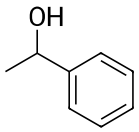
To a solution of the aryl methyl ketone (0.50 mmol, 1.0 equiv) in MeOH (2.5 mL) at 0 °C was added $NaBH_4$ (28.4 mg, 0.75 mmol, 1.5 equiv) in one portion. The mixture was stirred at 0 °C for 30 min (complete conversion by TLC), then quenched with saturated aqueous NH_4Cl (5 mL) and extracted with EtOAc (3 × 10 mL). The combined organic layers were washed with brine (10 mL), dried over Na_2SO_4, filtered and concentrated under reduced pressure.



1a–f → **2a–f** 89–96%

Scheme S1. Reduction of aryl methyl ketones 1a–1f (R = aryl)

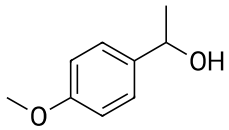
1-Phenylethan-1-ol (2a).



Prepared according to General Procedure A from acetophenone (**1a**, 60.1 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 4:1).

Colorless oil (57.4 mg, 94%). R_f = 0.23 (n-hexane/EtOAc 4:1). **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.39 – 7.34 (m, 4H), 7.30 – 7.27 (m, 1H), 4.90 (q, J = 6.5 Hz, 1H), 1.96 (s, 1H), 1.50 (d, J = 6.5 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 146.0, 128.6, 127.6, 125.5, 70.5, 25.3. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for $C_8H_{10}NaO$ 145.0624; found 145.0625. Spectral data matched those reported in the literature.[3,4]

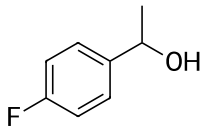
1-(4-Methoxyphenyl)ethan-1-ol (2b).



Prepared according to General Procedure A from 4′-methoxyacetophenone (**1b**, 75.1 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 3:1).

Colorless oil (69.2 mg, 91%). R_f = 0.10 (n-hexane/EtOAc 4:1). **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.34 – 7.28 (m, 2H), 6.92 – 6.86 (m, 2H), 4.87 (qd, J = 6.4, 3.4 Hz, 1H), 3.81 (s, 3H), 1.69 (d, J = 3.5 Hz, 1H), 1.49 (d, J = 6.4 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 159.2, 138.1, 126.8, 114.0, 70.2, 55.5, 25.2. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for $C_9H_{12}NaO_2$ 175.0730; found 175.0728. Spectral data matched those reported in the literature.[3–5]

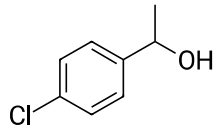
1-(4-Fluorophenyl)ethan-1-ol (2c).



Prepared according to General Procedure A from 4′-fluoroacetophenone (**1c**, 69.1 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 4:1).

Colorless oil (67.3 mg, 96%). R_f = 0.13 (n-hexane/EtOAc 4:1). **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.39 – 7.30 (m, 2H), 7.08 – 6.98 (m, 2H), 4.90 (q, J = 6.4 Hz, 1H), 1.74 (s, 1H), 1.48 (d, J = 6.4 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 162.3 (d, J = 244.9 Hz), 141.7 (d, J = 3.2 Hz), 127.2 (d, J = 8.1 Hz), 115.4 (d, J = 21.4 Hz), 70.0, 25.5. **^{19}F{^{1}H} NMR** (376 MHz, $CDCl_3$) δ −115.5. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for C_8H_9FNaO 163.0530; found 163.0532. Spectral data matched those reported in the literature.[3–5]

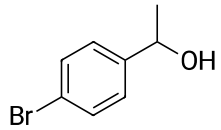
1-(4-Chlorophenyl)ethan-1-ol (2d).



Prepared according to General Procedure A from 4′-chloroacetophenone (**1d**, 77.3 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 4:1).

Pale yellow oil (72.8 mg, 93%). R_f = 0.16 (n-hexane/EtOAc 4:1). **^{1}H NMR** (400 MHz, $CDCl_3$) δ 7.35 – 7.27 (m, 4H), 4.94 – 4.84 (m, 1H), 1.79 (d, J = 3.6 Hz, 1H), 1.48 (d, J = 6.5 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 144.4, 133.2, 128.8, 126.9, 69.9, 25.4. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for C_8H_9ClNaO 179.0234; found 179.0233. Spectral data matched those reported in the literature.[4,5]

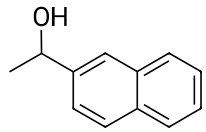
1-(4-Bromophenyl)ethan-1-ol (2e).



Prepared according to General Procedure A from 4′-bromoacetophenone (**1e**, 99.5 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 4:1).

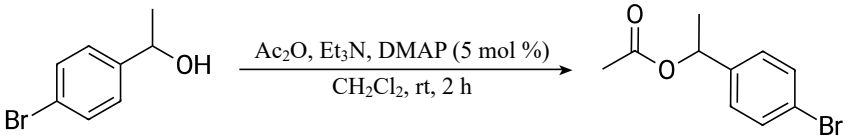
Colorless oil (95.5 mg, 95%). R_f = 0.17 (n-hexane/EtOAc 4:1). **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.47 (d, J = 8.5 Hz, 2H), 7.25 (d, J = 8.5 Hz, 2H), 4.87 (qd, J = 6.4, 3.6 Hz, 1H), 1.77 (d, J = 3.7 Hz, 1H), 1.47 (d, J = 6.5 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 144.9, 131.7, 127.3, 121.3, 70.0, 25.4. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for C_8H_9BrNaO 222.9729; found 222.9727. Spectral data matched those reported in the literature.[3–5]

1-(Naphthalen-2-yl)ethan-1-ol (2f).



Prepared according to General Procedure A from 2′-acetonaphthone (**1f**, 85.1 mg, 0.50 mmol). The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 3:1).

Colorless solid (76.6 mg, 89%). R_f = 0.50 (n-hexane/EtOAc 3:1). **^{1}H NMR** (600 MHz, $CDCl_3$) δ 7.86 – 7.80 (m, 4H), 7.53 – 7.44 (m, 3H), 5.07 (q, *J* = 6.4 Hz, 1H), 1.98 (br s, 1H), 1.59 (d, *J* = 6.4 Hz, 3H). **^{13}C NMR** (150 MHz, $CDCl_3$) δ 143.3, 133.4, 133.1, 128.4, 128.1, 127.8, 126.3, 125.9, 123.94, 123.93, 70.6, 25.3. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for $C_{12}H_{12}NaO$ 195.0780; found 195.0781. Spectral data matched those reported in the literature.[6]



2e **3** 88%

Scheme S2. Acetylation of 1-(4-bromophenyl)ethan-1-ol

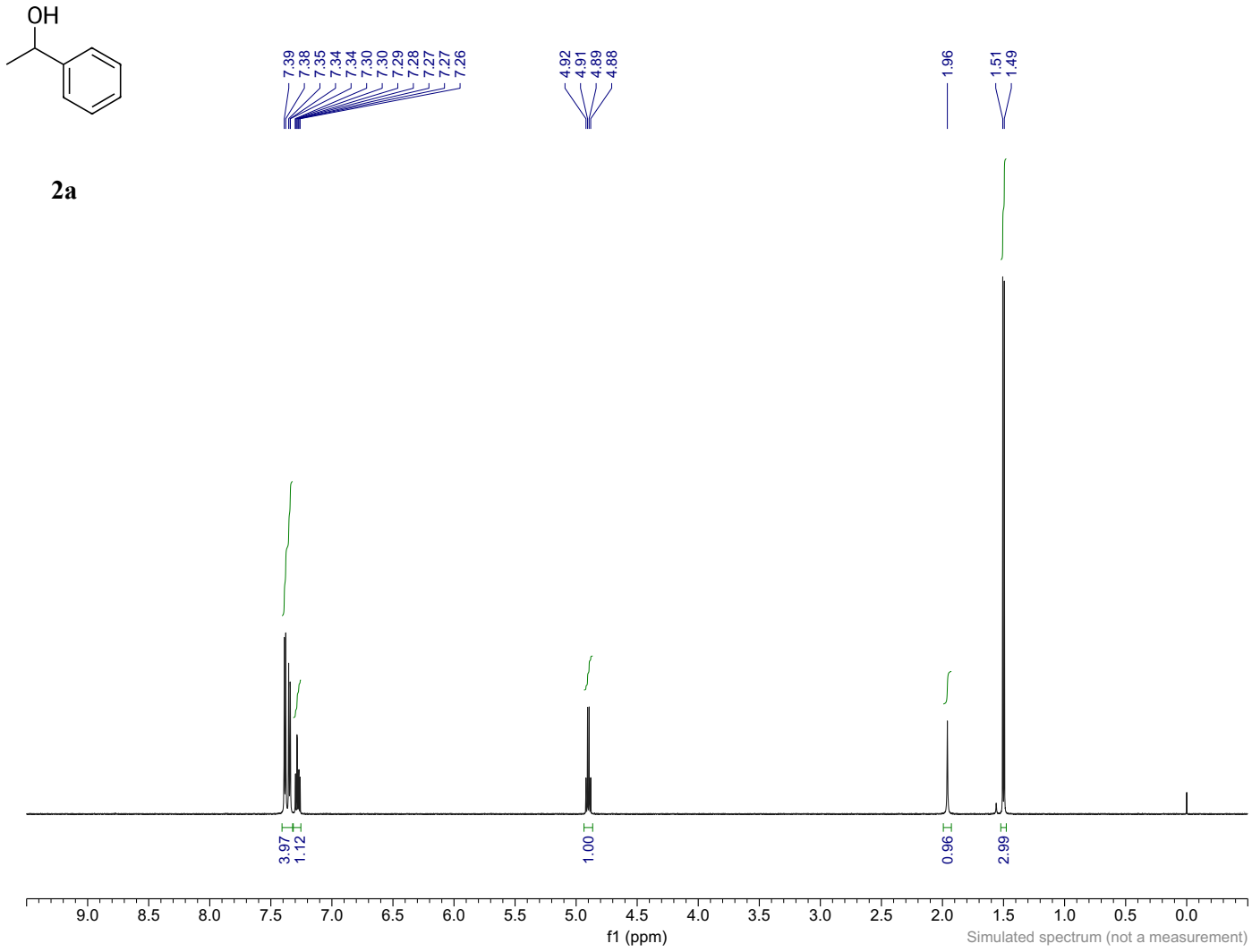
1-(4-Bromophenyl)ethyl acetate (3).

To a solution of alcohol **2e** (80.4 mg, 0.40 mmol, 1.0 equiv) in CH_2Cl_2 (2.0 mL) were added Et_3N (84 µL, 0.60 mmol, 1.5 equiv), DMAP (2.4 mg, 0.020 mmol, 5 mol %) and Ac_2O (57 µL, 0.60 mmol, 1.5 equiv) at room temperature. After 2 h (complete conversion by TLC) the mixture was diluted with CH_2Cl_2 (10 mL), washed with 1 M HCl (5 mL), saturated aqueous $NaHCO_3$ (5 mL) and brine (5 mL), dried over Na_2SO_4, filtered and concentrated under reduced pressure. The crude product was purified by flash column chromatography (SiO_2, n-hexane/EtOAc 10:1).

Colorless oil (85.6 mg, 88%). **^{1}H NMR** (500 MHz, $CDCl_3$) δ 7.49 – 7.45 (m, 2H), 7.24 – 7.20 (m, 2H), 5.82 (q, *J* = 6.6 Hz, 1H), 2.07 (s, 3H), 1.51 (d, *J* = 6.6 Hz, 3H). **^{13}C NMR** (126 MHz, $CDCl_3$) δ 170.5, 141.1, 132.0, 128.2, 122.1, 72.0, 22.5, 21.6. **HRMS** (ESI) *m/z*: $[M+Na]^+$ calcd for $C_{10}H_{11}BrNaO_2$ 264.9835; found 264.9837. Spectral data matched those reported in the literature.[7]

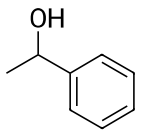
Spectra

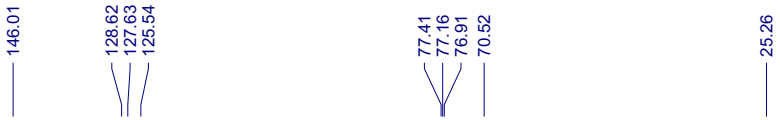
2a — ^{1}H NMR (500 MHz, $CDCl_3$)



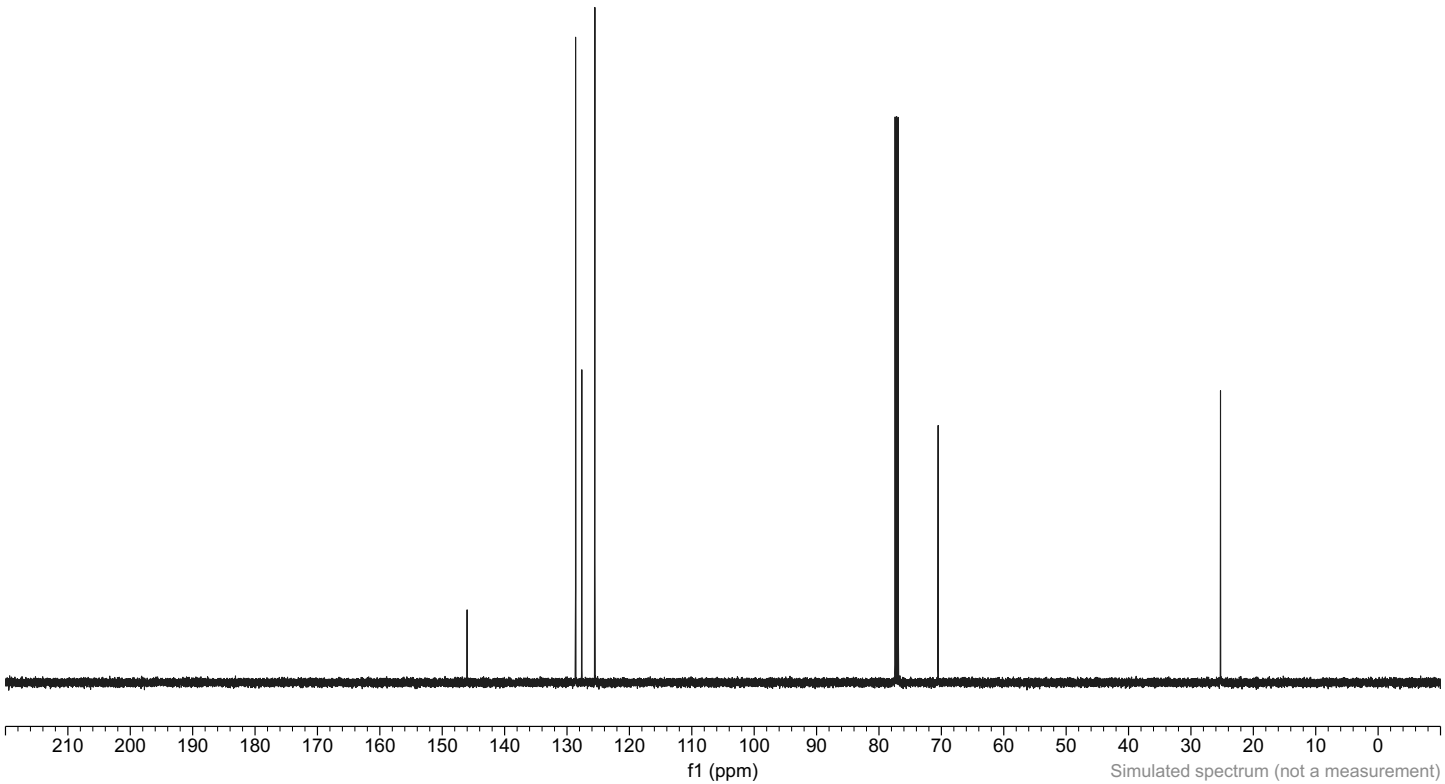
2a

2a — ^{13}C NMR (126 MHz, $CDCl_3$)

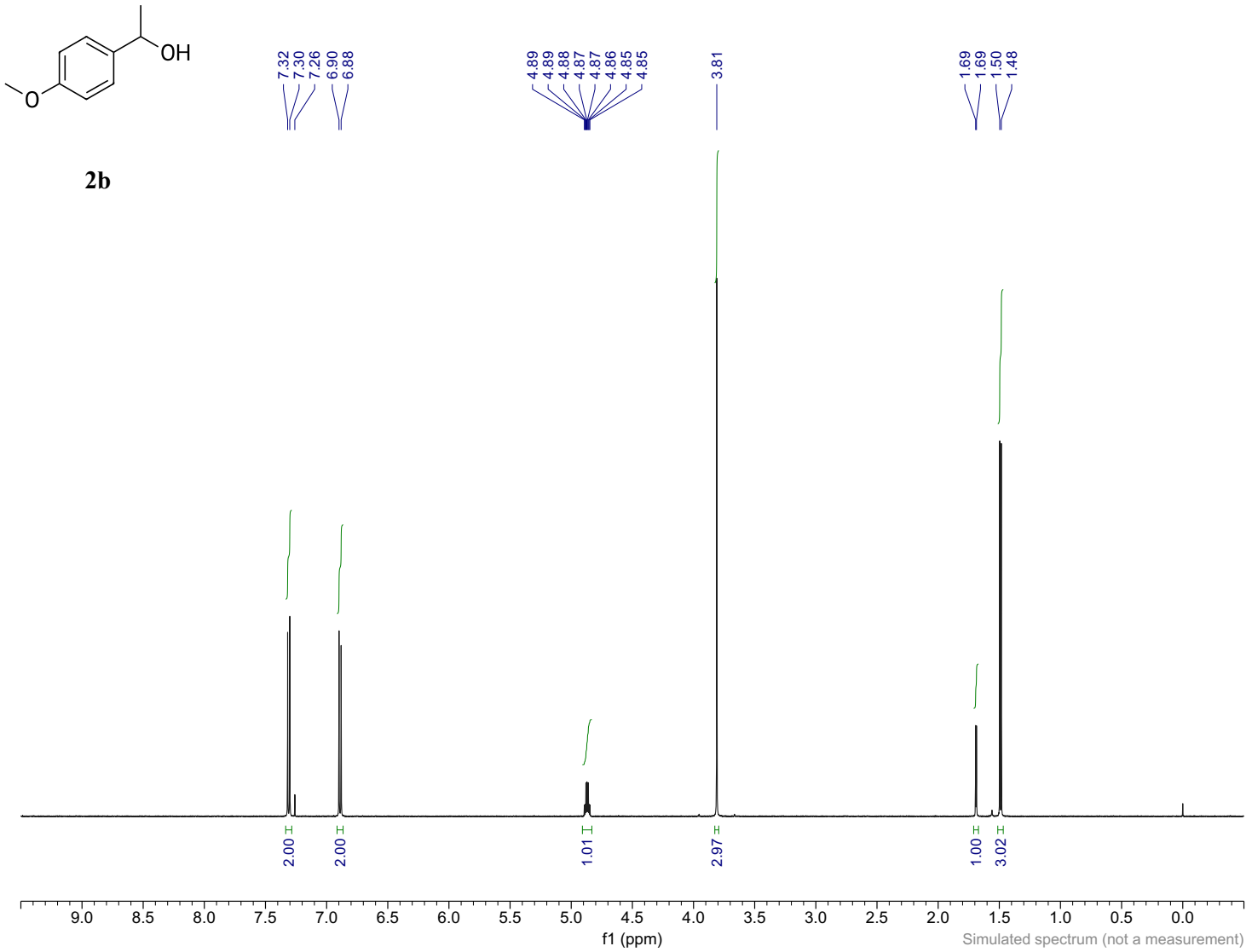




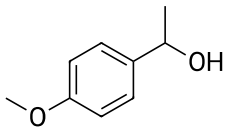
2a



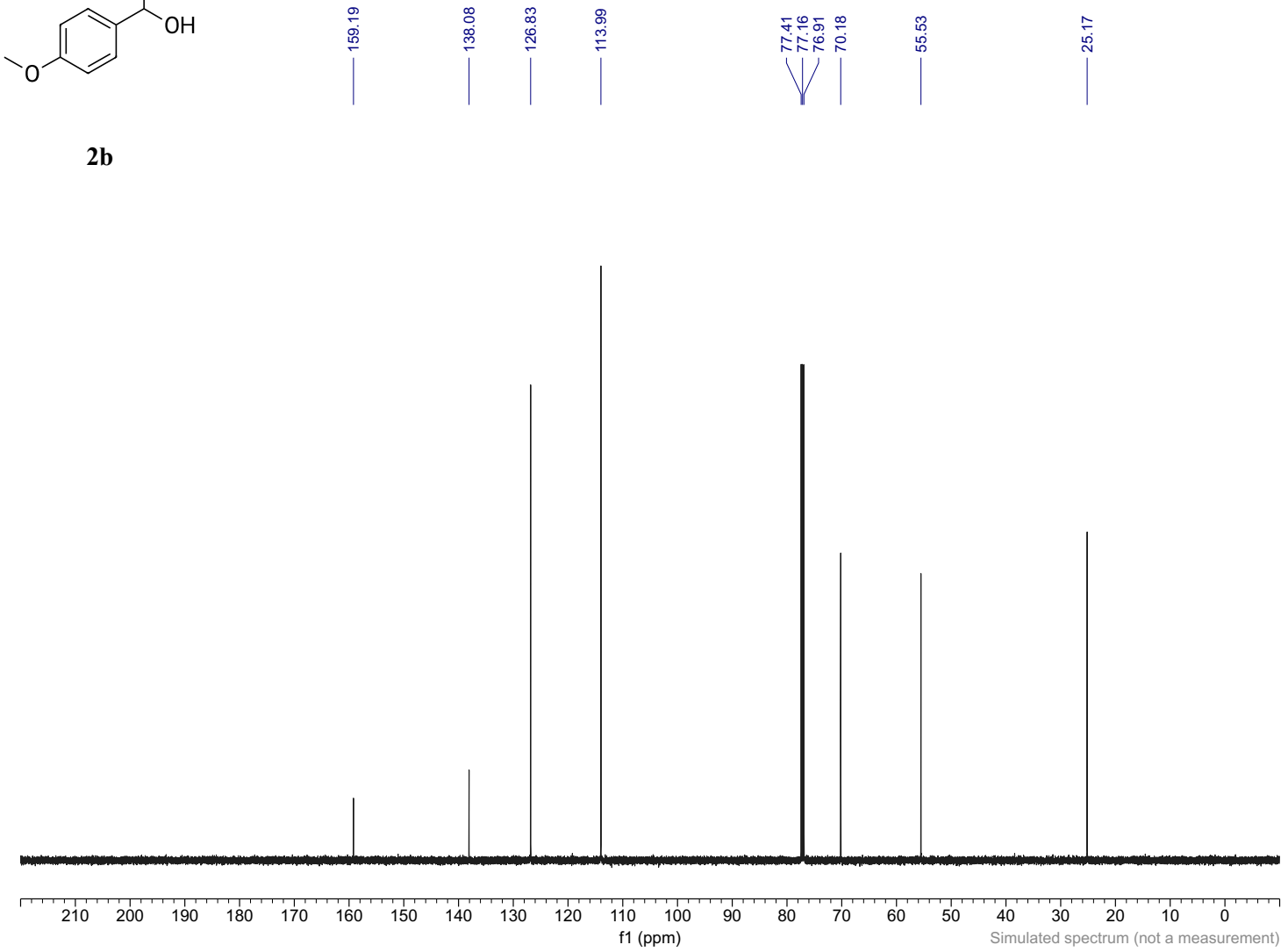
2b — ^{1}H NMR (500 MHz, $CDCl_3$)



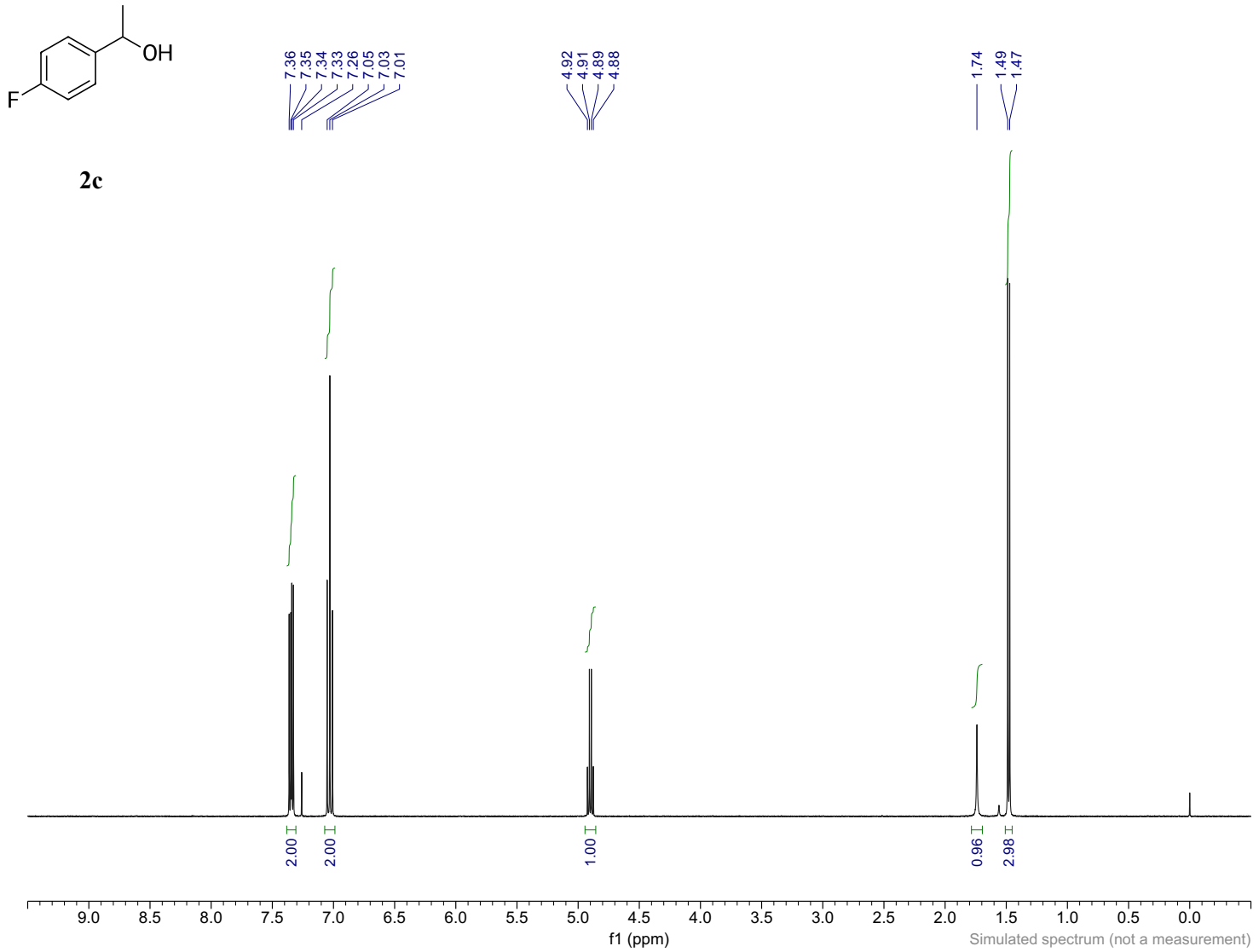
2b — ^{13}C NMR (126 MHz, $CDCl_3$)



2b

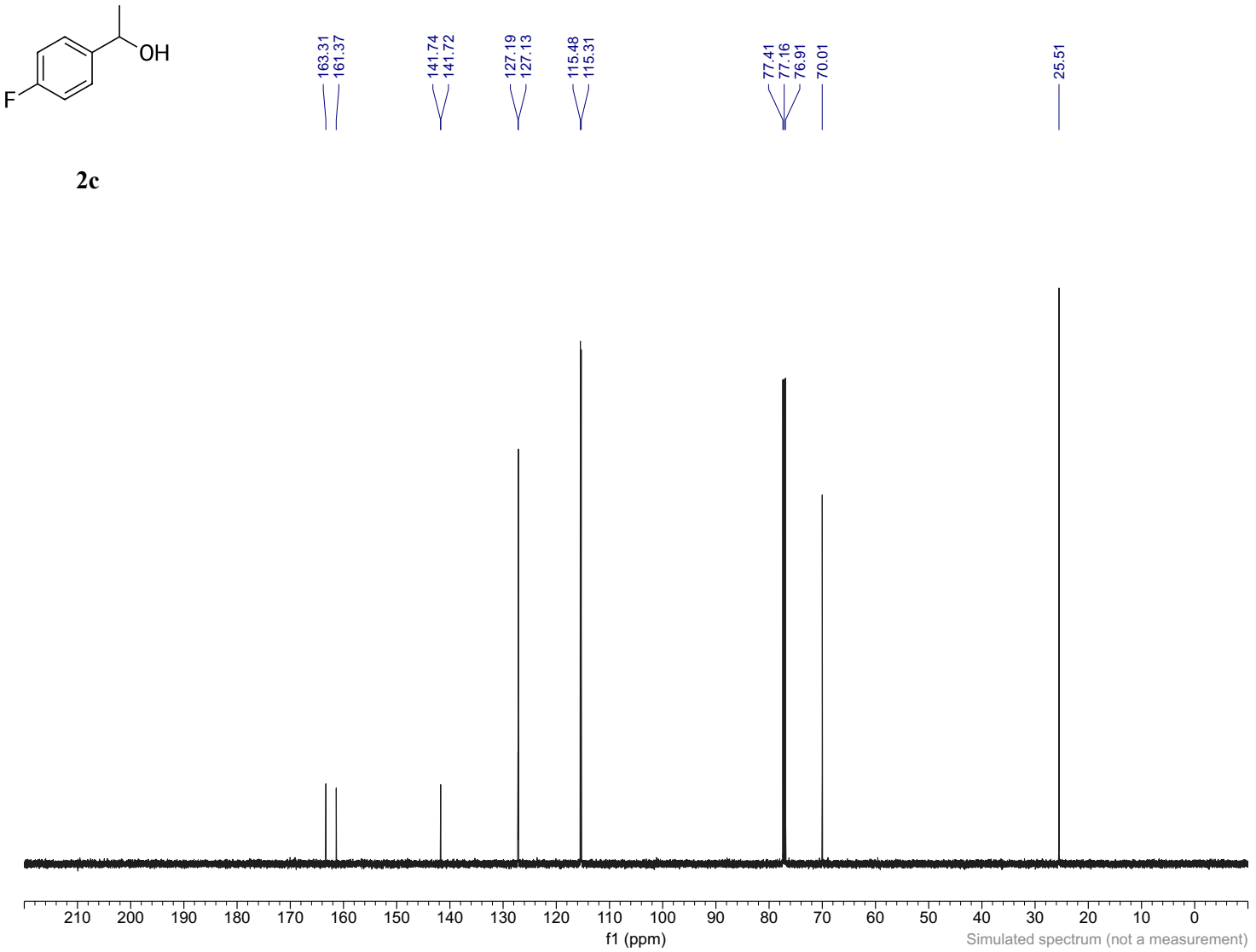


2c — ^{1}H NMR (400 MHz, $CDCl_3$)



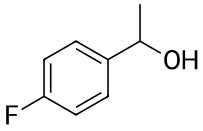
2c

2c — ^{13}C NMR (126 MHz, $CDCl_3$)

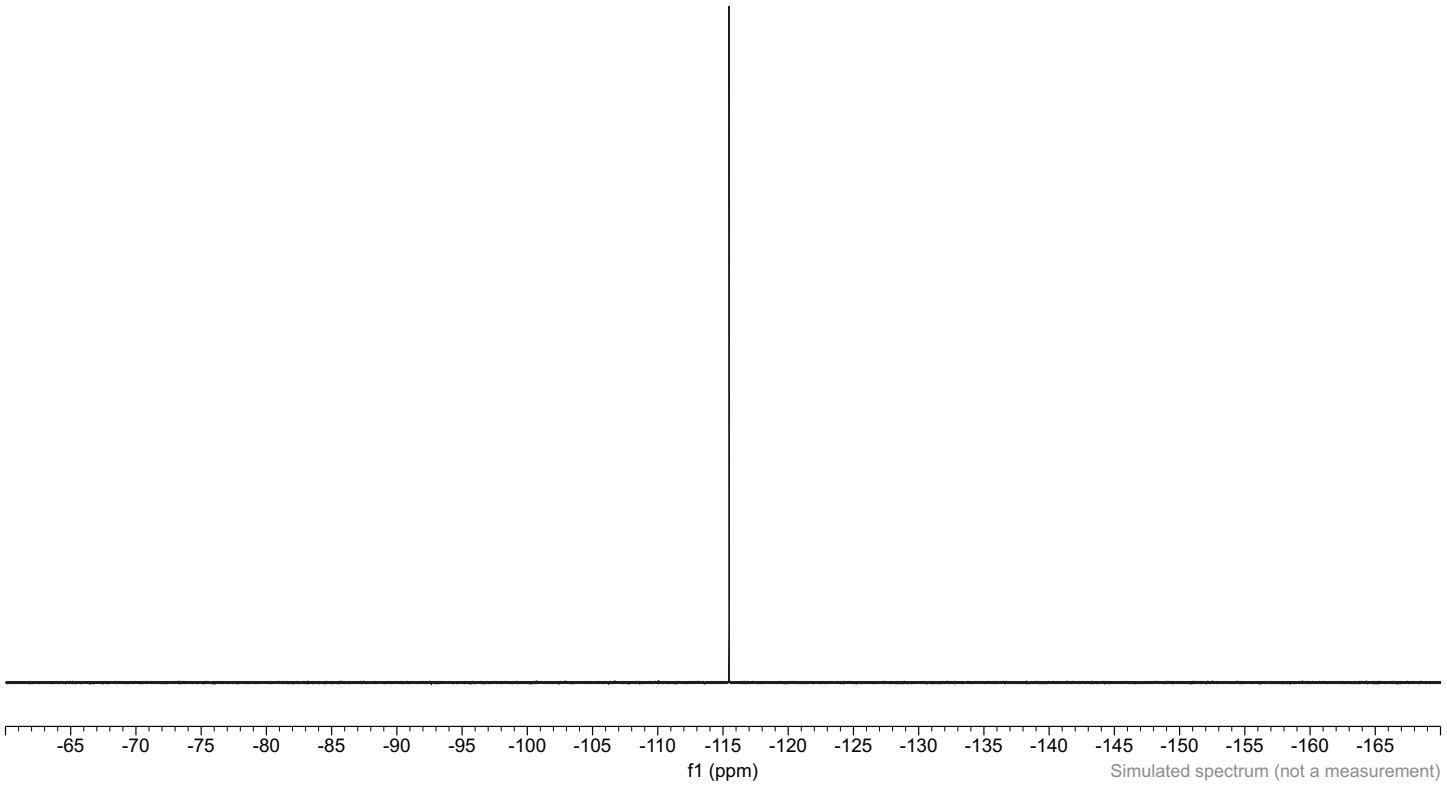


2c

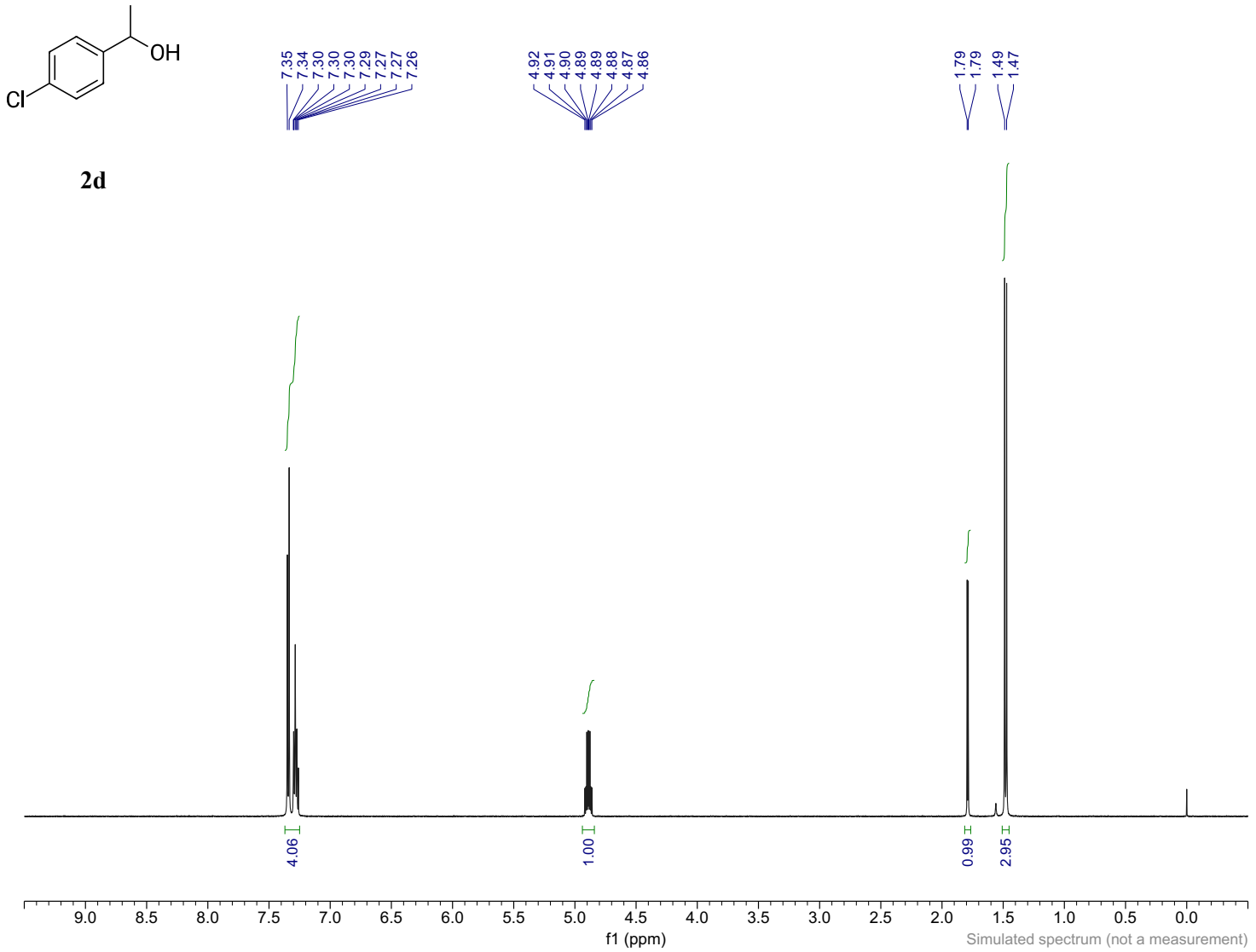
2c — $^{19}F\{^{1}H\}$ NMR (376 MHz, $CDCl_3$)



2c

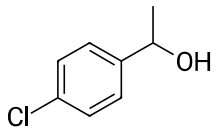


2d — ^{1}H NMR (400 MHz, $CDCl_3$)

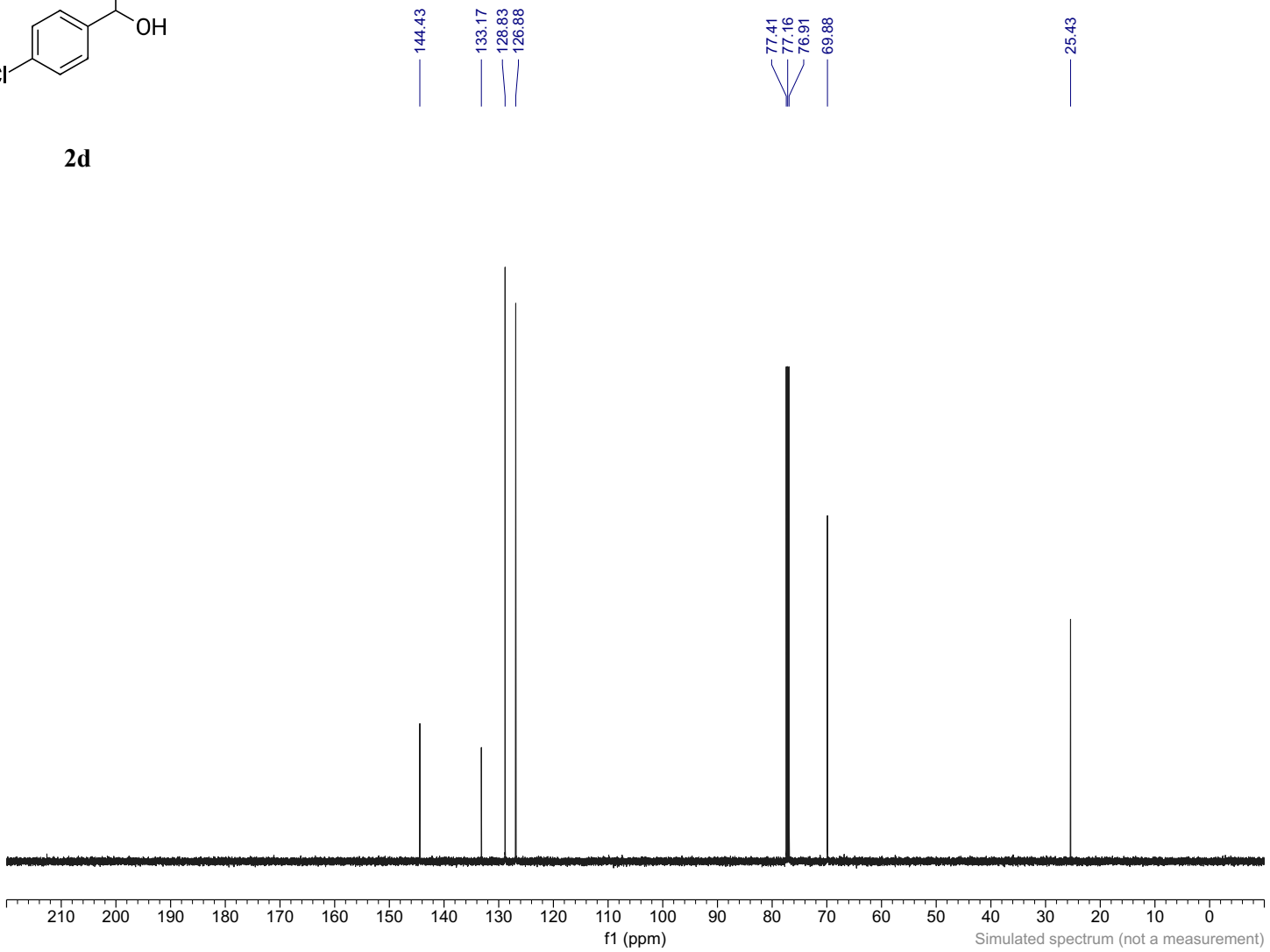


2d

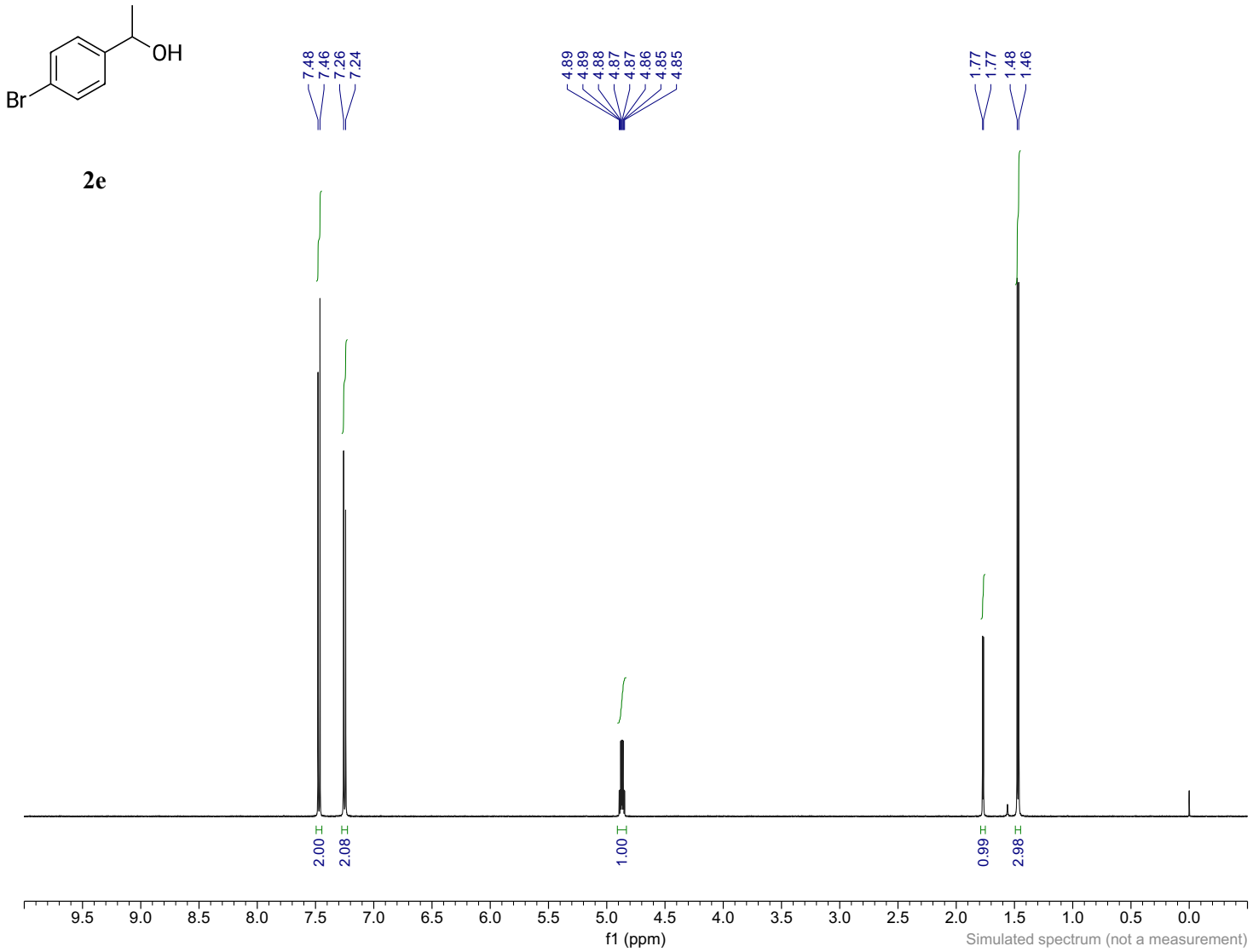
2d — ^{13}C NMR (126 MHz, $CDCl_3$)



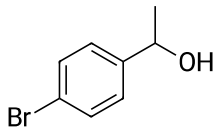
2d



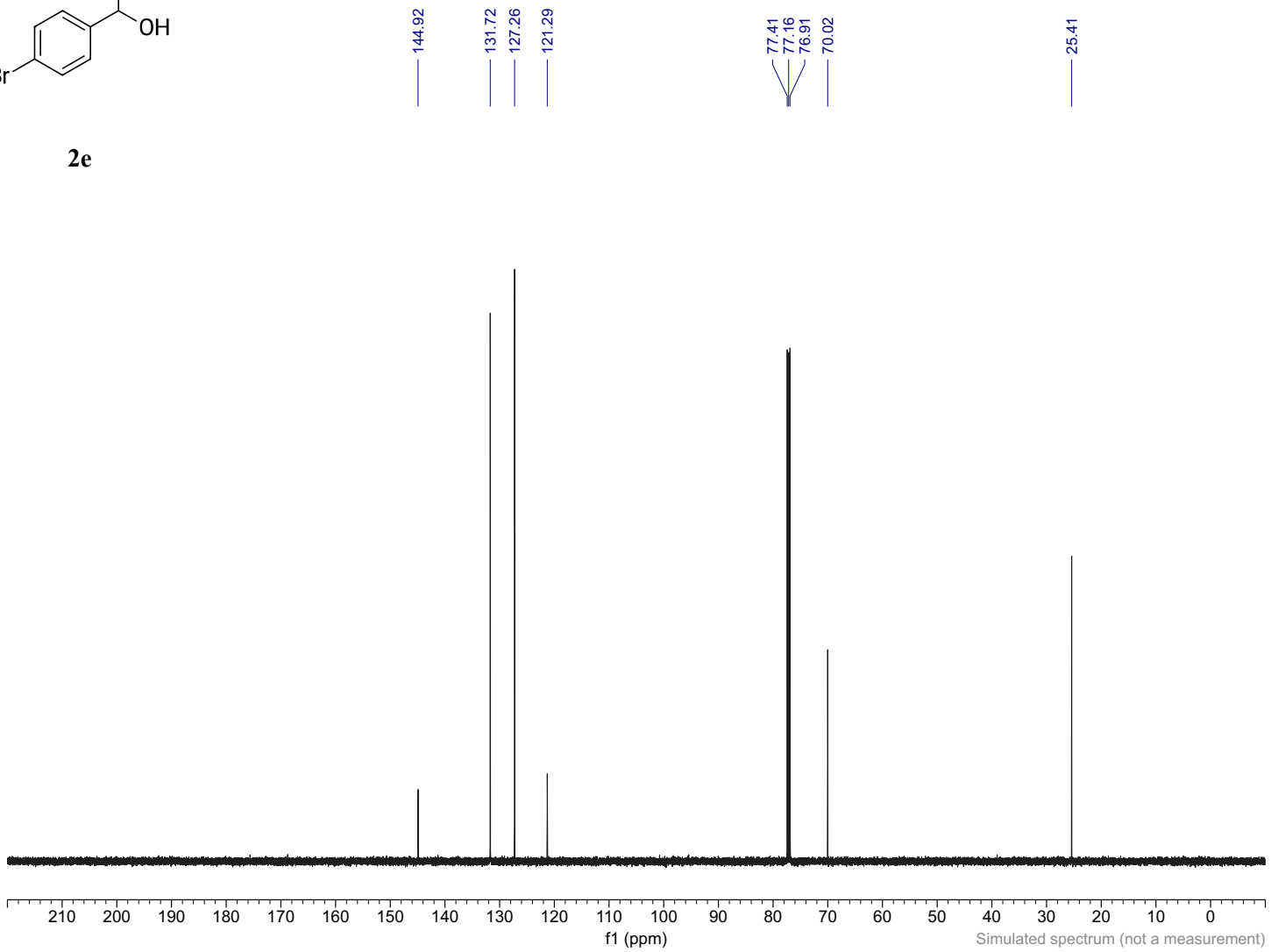
2e — ^{1}H NMR (500 MHz, $CDCl_3$)



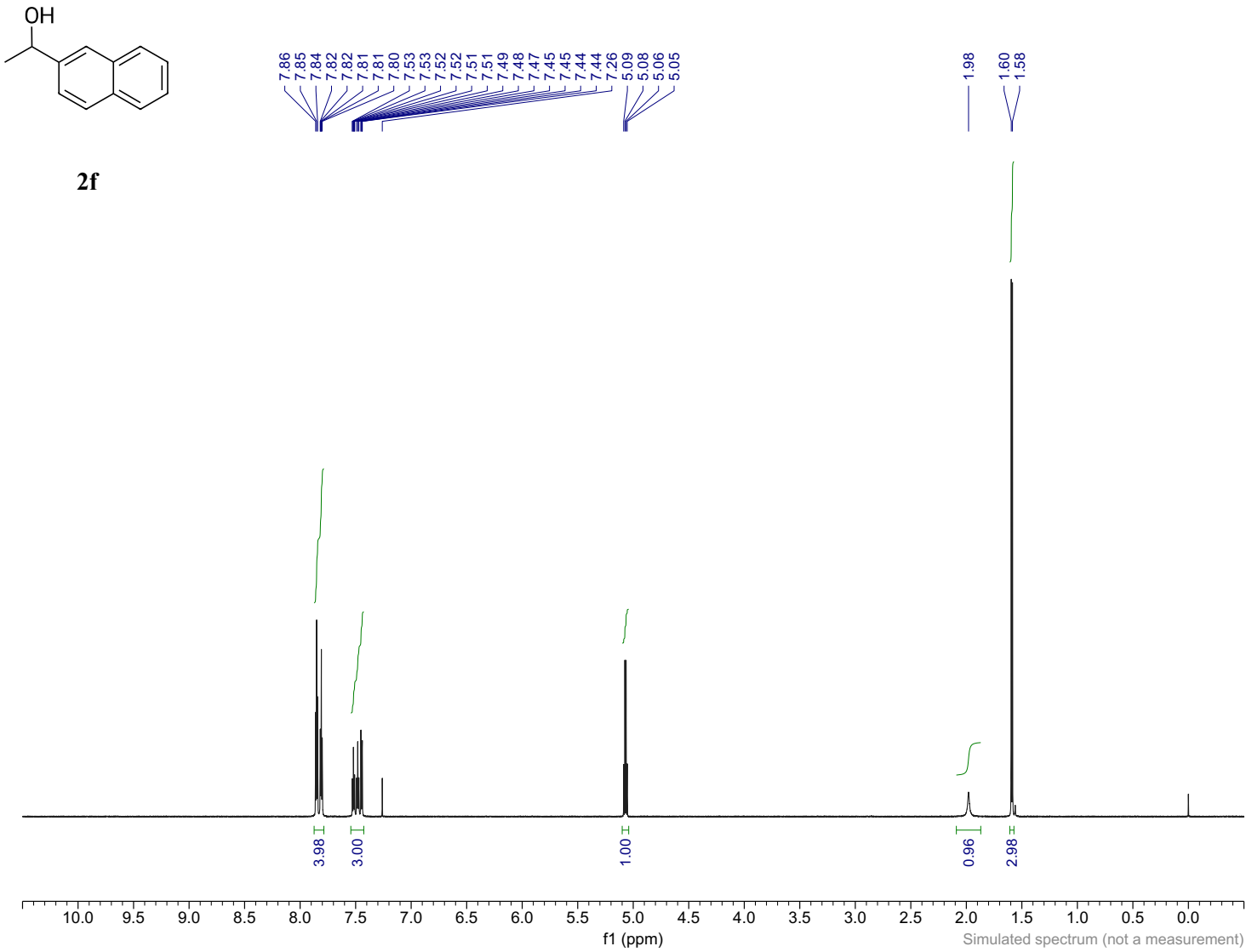
2e — ^{13}C NMR (126 MHz, $CDCl_3$)



2e

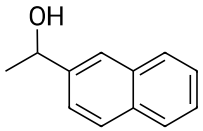


2f — ^{1}H NMR (600 MHz, $CDCl_3$)

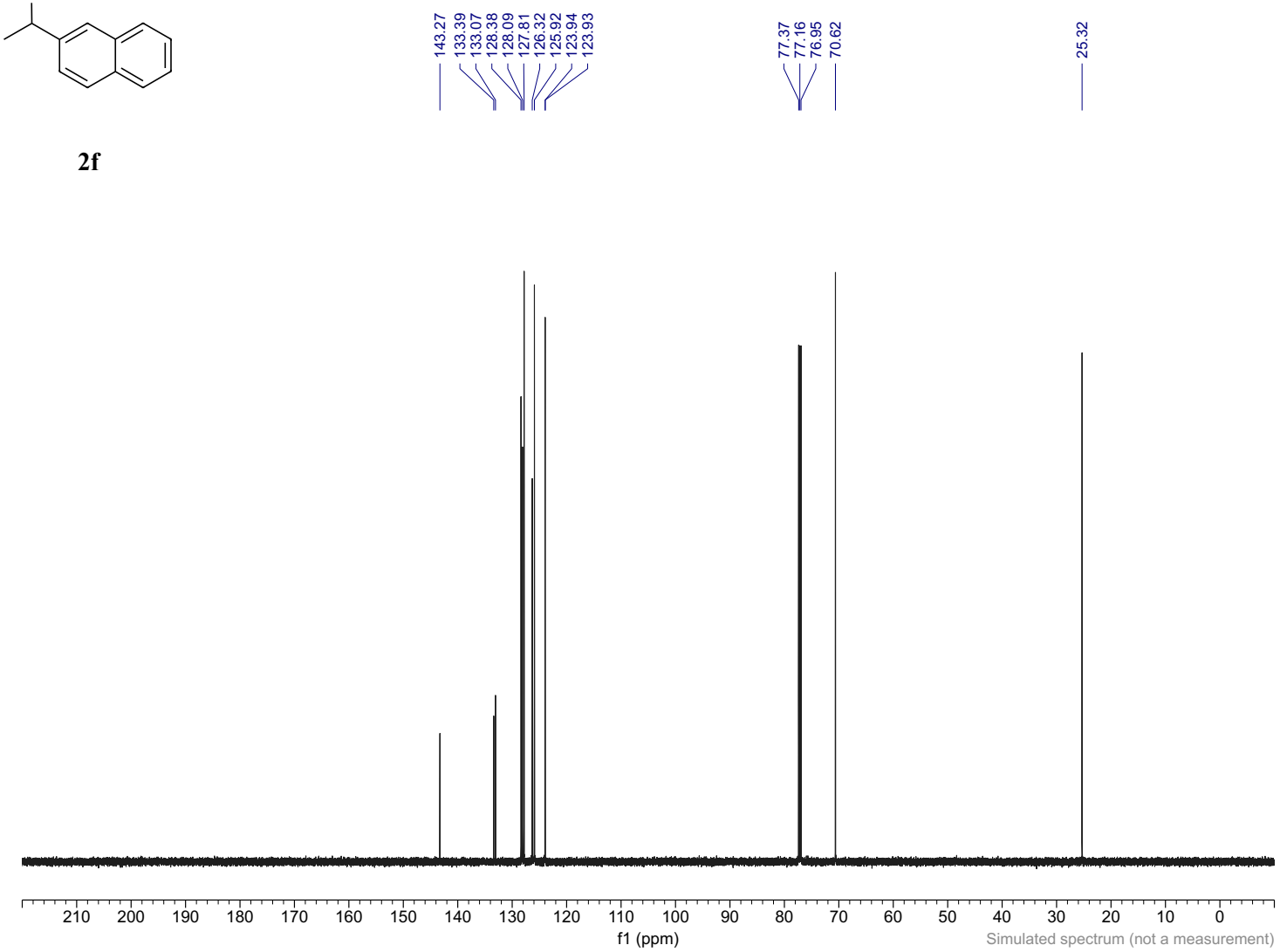


2f

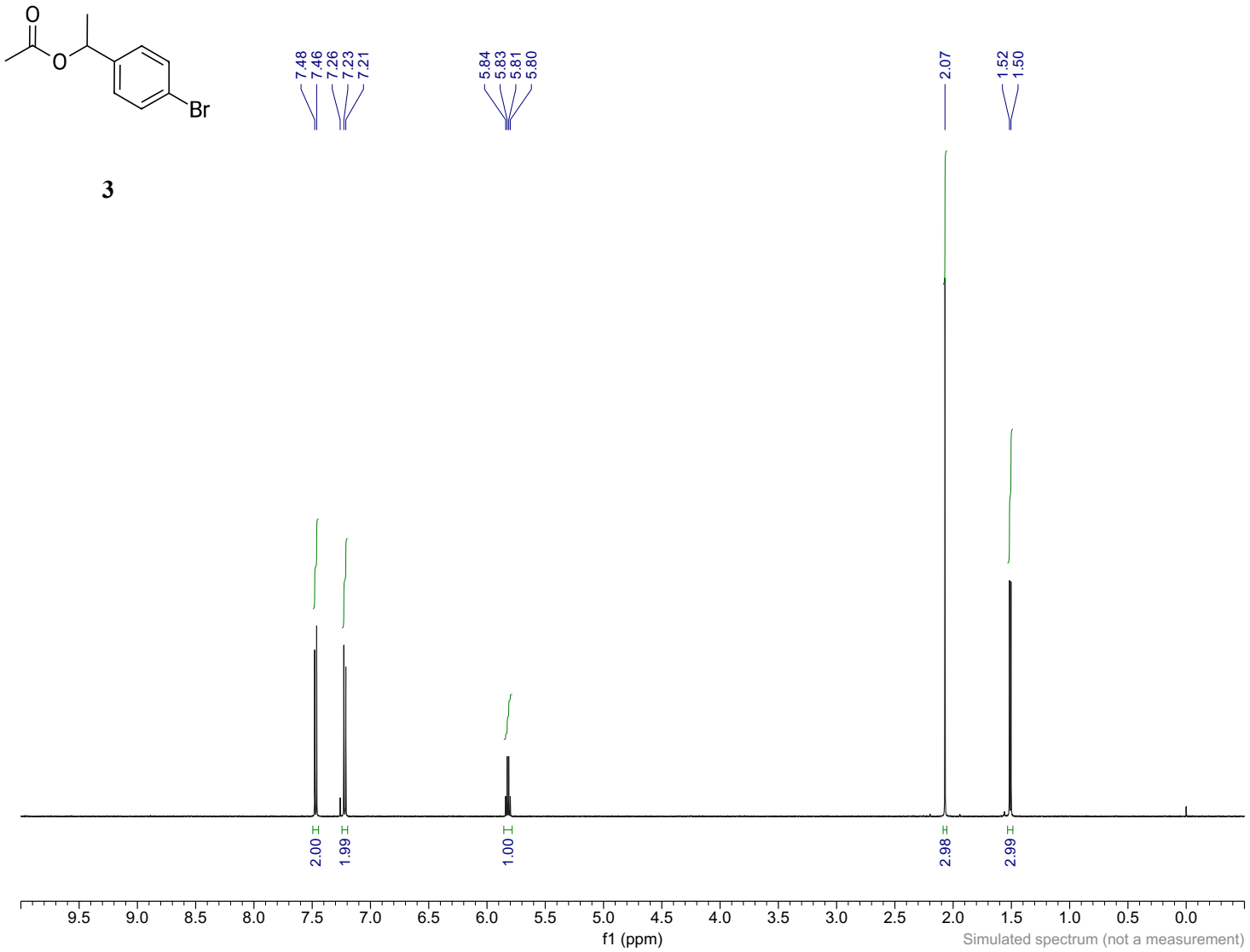
2f — ^{13}C NMR (150 MHz, $CDCl_3$)



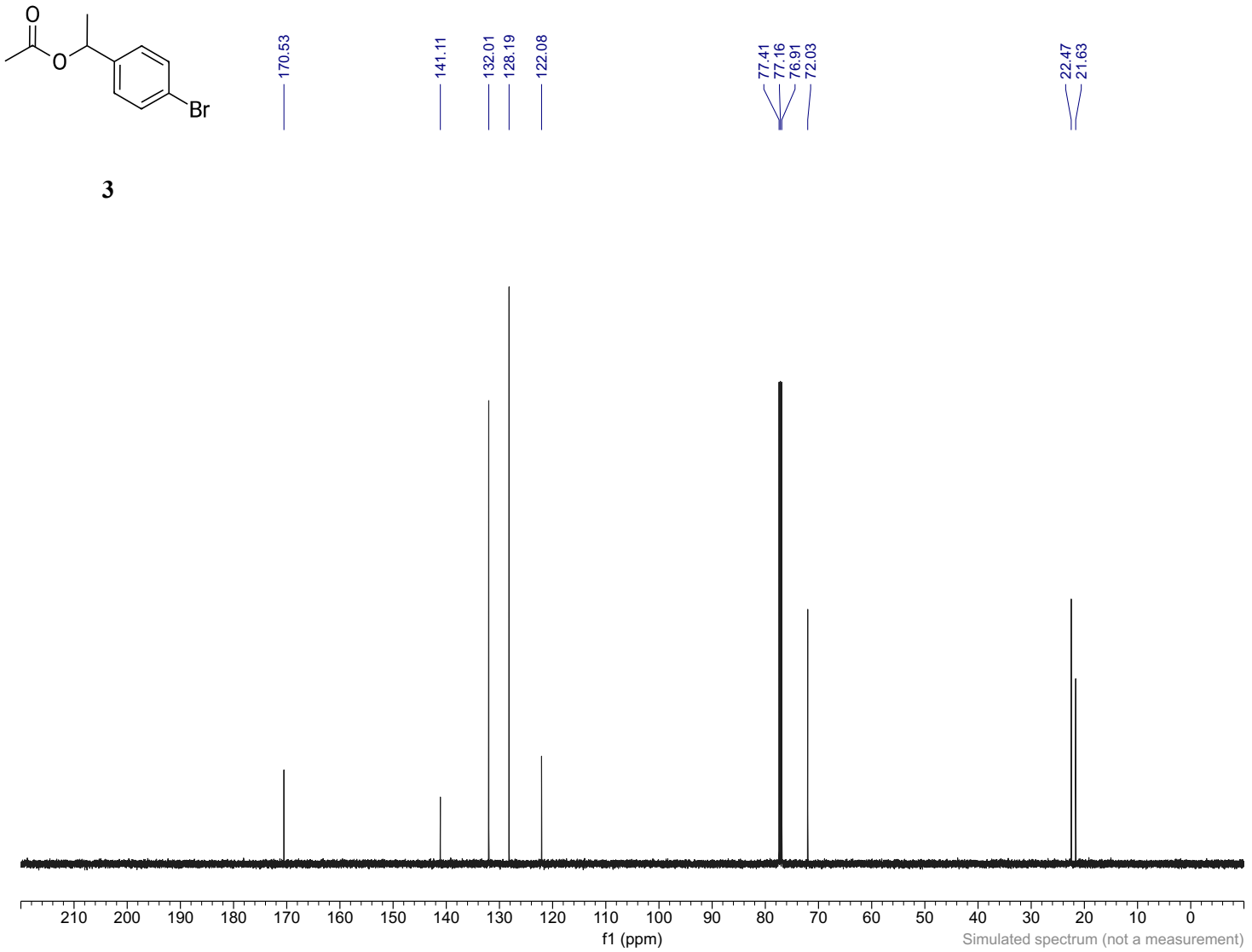
2f



3 — ^{1}H NMR (500 MHz, $CDCl_3$)



3 — ^{13}C NMR (126 MHz, $CDCl_3$)



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References

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